

Challenge (Habanero)

- Q1.** Solve the system: $x + y = 4$, $x^2 + y^2 = 16$. What is the value of x ?
(A) 0
(B) 2
(C) 4
(D) 6
(E) 8
- Q2.** A recipe for making cookies requires $2x$ cups of flour and $x + 3$ cups of sugar. If the total amount of flour and sugar is 14 cups, what is the value of x ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q3.** A building has a height of x meters. A stone is thrown from the top of the building with an initial velocity of $20 \cdot x$ m/s. What is the time it takes for the stone to reach the ground if $x = 10$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q4.** A construction company is building a rectangular foundation with a length of $x + 5$ meters and a width of $x - 2$ meters. If the area of the foundation is 48 square meters, what is the value of x ?
(A) 6
(B) 7
(C) 8
(D) 9
- (E) 10
- Q5.** Solve the system: $y = 2x - 3$, $y = x^2 - 4$. What is the y -coordinate of the intersection point?
(A) -5
(B) -3
(C) -1
(D) 1
(E) 3
- Q6.** A baker is making a cake that requires x cups of flour and $2x + 1$ cups of sugar. If the total amount of flour and sugar is 21 cups, what is the value of x ?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q7.** A architect designs a bridge with a length of $x + 10$ meters and a height of $x - 5$ meters. If the area of the bridge is 60 square meters, what is the value of x ?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16
- Q8.** Solve the system: $x^2 + y^2 = 25$, $y = x - 2$. What is the value of y ?
(A) -5
(B) -3
(C) -1
(D) 1
(E) 3

Open Ended Questions (FRQ)

- Q9.** A contractor is building a house with a rectangular foundation. The length of the foundation is $x + 2$ meters and the width is $x - 1$ meter. If the area of the foundation is 50 square meters, find the value of x and the dimensions of the foundation.
- Q10.** A chef is making a recipe that requires x cups of flour and $3x + 2$ cups of sugar. If the total amount of flour and sugar is 30 cups, find the value of x and the total amount of each ingredient.

Chef's Tips

- Review the basics of linear and quadratic equations before starting the test
- Emphasize the importance of reading the questions carefully and understanding the context
- Encourage students to use graphs and tables to visualize the systems of equations